

F20FO Digital Forensics Assessed Coursework 2

Digital Forensics Investigation

1. Coursework Description

You are employed as a Digital Forensics Analyst for a Cyber Security company. You have been requested by your line manager to work on a case, which involves a piece of evidence recovered from a suspect and there is no case background for the moment. You have been assigned to create a forensically sound image of the seized digital evidence on the storage device. You are then expected to perform a forensic analysis of the image. The digital evidence will be available from the F20FO course page on Canvas.

You are required to correctly process the digital evidence, maintain continuity and integrity of that evidence, and take notes at every step of your investigation to ensure continuity of the evidence. Your tasks are:

- To create a verifiable forensic image of the media. You should complete the imaging twice, once using FTK Imager, and then using a forensic tool of your choice. In your report you should discuss and compare the process and results for creating the two images.
- To perform a forensic analysis of the forensic image(s) you generated, use Autopsy and a second forensic tool of your choice. You should analyse the contents of the image and identify files that would be of interest for the investigation. In your report you should discuss and compare the process and results obtained from the two analysis tools.
- You need to document the forensic imaging process, chain of custody and your analysis of the files of interest recovered in a short technical report. Your report should also include a section on any issues faced during the analysis process and you need to record the files that were recovered.
- You must make use of the virtual digital forensics labs you created in coursework 1 (CW1) and, if required, add additional forensic tools to the labs. It is up to the group members to decide how best to collaborate and use their labs to complete the coursework.

You should make use of the digital forensics concepts, the tools and the techniques covered in the lecture materials and lab sessions for this course. Support will be available in weeks 10 and 11 in the lab sessions for the coursework. To demonstrate your work, your group will need to participate in a demonstration viva with your course lecturer to be arranged locally, showcasing the imaging and analysis. Marks will be awarded as follows:

Section	Description	Marks
Forensic imaging	Creation of the forensic images.	7
Forensic Analysis	Thoroughness of the analysis and files recovered, while maintaining chain of custody. Use of labs.	15
Demonstration Ability	Your ability to demonstrate and explain your work.	8
Technical Report	Documenting DF analysis, and quality of the overall report.	5
Individual Report	Your individual contributions and key challenges or successes.	5

2. Group Work

This coursework will be conducted in groups of 2 students. You will form your own groups; however, the teaching team will make final adjustments if necessary. Group members must be from the same cohort and campus. Group members must arrange between themselves on how to distribute the workload. If there are issues within groups, please contact your respective course tutor.

Most of the marks will be allocated as a group component; however, this group mark will be modified for each member according to their individual contribution, measured by the individual report and the peer review form.

You must register yourself to a group on Canvas by week 10 (Select *People* section, then *CW2* tab). The lecturer will complete the group formation if:

- You are not registered in a group; or
- Less than 2 students are registered to a group.

3. Report Style

3.1 Group report

Your report needs to include the following sections:

- Group number, campus, cohort and full student names
- An introduction to the report
- The core sections of the report
 - Forensic imaging process
 - Forensic analysis
- A single paragraph concluding the report
- Appendix for the chain of custody (referring to key stages in the core sections of your report)

The report should not exceed 7 A4 pages (not including Title, Appendix or Reference pages). The report should not be written in the first person (singular or plural). You are required to use diagrams or tables where appropriate. The contents of the technical report will be used to assess the forensic imaging and analysis tasks.

3.1 Individual report

Your individual report should have a maximum of 2 A4 pages (not including references) and contain:

- Your name, campus, cohort and group number.
- A **personal reflection** about your contribution to this project. Using precise examples, highlight key challenges (technical issues, knowledge gap, group work problems, etc.), how you personally addressed them, and what you have learned from them. You can also reflect on the positive elements and how you would reimplement them or improve upon them in the future.

4. Submission

One member of the group should submit an electronic version of the group report via Canvas. You need to save a copy of your forensics image(s) in one of the team member's student OneDrive or their private repository, note that the storage size of the images can be quite large. Share the link to the forensic image(s) with your respective course tutor (Mike Just, Ryad Soobhany) and put the shared link in the group report above the Introduction section. It is the responsibility of the group to ensure that access to the shared files is granted to their course tutor.

Each student must submit their own individual report via Canvas.

The demonstration will be scheduled by your course tutor and arranged per campus. Your group needs to showcase the forensic imaging and analysis of your coursework for a maximum of 5 minutes, which will be followed by questions from your course tutor.

The deadline for submitting the report is **Wednesday 1st April 2026 at 2359**. Late submissions will be subject to the normal penalties as defined on the next page.

If you have any questions or queries about the assessment, please do not hesitate to contact m.just@hw.ac.uk (Edinburgh) or r.soobhany@hw.ac.uk (Dubai).

Late Submission

The University recognises that, on occasion, students may be unable to submit coursework on the submission date or be unable to present their work on the submission date. In these cases, the University's Submission of Coursework Policy outlines are:

- No individual extensions are permitted under any circumstances.
- Standard 30% deduction from the mark awarded (maximum of five working days).
- In the case where a student submits coursework up to five working days late, and the student has valid mitigating circumstances, the mitigating circumstances policy will apply, and appropriate mitigation will be applied.
- Any coursework submitted after five working days of the set submission date shall be automatically awarded a no grade with no formative feedback provided.

5. Notes

- Coursework reports must be written in your own words. If some text or diagrams in the coursework has been taken from other sources, these sources must be properly referenced and should be open for use. The link to the University's policy on Plagiarism is:

<https://www.hw.ac.uk/students/studies/examinations/plagiarism.htm>

- Failure to reference work that has been obtained from other sources or to copy the words of another student is plagiarism, and if detected, this will be reported to the School's Discipline Committee. If a student is found guilty of plagiarism, the penalty could involve voiding the course.
- You must never give hard or soft copies of your coursework reports to another student. You must always refuse any request from another student for a copy of your report.
- Sharing a coursework report with another student is collusion, and if detected, this will be reported to the School's Discipline Committee. If found guilty of collusion, the penalty could involve voiding the course.
- You can expect to receive formative feedback on the coursework within 15 working days after the submission deadline.
- You may use GenAI to assist you to develop ideas, structures (e.g. headings, bullet points, outline plans), identify potential research themes or journal articles in your preparation of your assessment, but no GenAI generated content is permitted in your assessment submission.

6. Marking Guidelines

Assessment Section	Below Threshold (F – E)	Threshold (E – C)	Above Threshold (C – A)
Forensic Imaging	(0.0 – 2.0) Unable to perform forensic imaging. Only one tool or method used for imaging. Limited or no knowledge of imaging process. Did not make use of DF lab. Limited comparison of the two imaging process and results.	(2.0 – 4.5) Two forensic tools or methods used with average explanation. Average knowledge of imaging process. Average use of DF lab with some additional tools. Some comparison of the two imaging process and results.	(4.5 – 7.0) Two forensic tools or methods used, with excellent explanation, and tool comparison. Very good knowledge of imaging process. Added relevant tools and made excellent use of DF lab. Excellent comparison of the two imaging process and results.

Forensic Analysis	(0.0 – 5.0) Missing or very limited forensic analysis. Chain of custody principles not followed properly. No or limited file recovery performed with no proper explanation. Did not make use of DF lab. Used one tool for analysis. Limited comparison of the process and analysis for the two tools.	(5.0 – 10.0) Average forensic analysis. Chain of custody principles followed but lacking quality for some steps. Good file recovery performed with average explanation. Average use of DF lab with some additional tools. Second tool with limited functionalities. Average highlighting of difference in tools (if any). Some comparison of the process and analysis for the two tools.	(10.0 – 15.0) Excellent forensic analysis performed. Chain of custody principles followed and all steps documented clearly. Complete file recovery performed with good explanation. Added relevant tools and made excellent use of DF lab. Second tool with advanced functionalities. Excellent comparison and highlighting of difference in tools (if any). Excellent comparison of the process and analysis for the two tools.
Demonstration	(0.0 – 2.0) Student(s) unable to demonstrate and explain their forensics imaging and analysis. Very limited knowledge of forensic process.	(2.0 – 5.0) Student(s) could demonstrate and explain their forensics imaging and analysis. Average knowledge of forensic process	(5.0 – 8.0) Student(s) able demonstrate and explain their forensics imaging and analysis confidently and in depth.
Technical report	(0.0 – 1.0) Missing sections and difficult to understand. No references and citations. No discussion of files recovered and tools used.	(1.0 – 3.0) All sections are available, and the report is comprehensible. Some references and citations used. Average discussion of files recovered and tools used. Some mention of selected issues.	(3.0 – 5.0) Care was taken to produce the report, including code formatting and appendices. Technical quality is shown in the report. References and citations used effectively and formatted correctly. Excellent and innovative process documentation and discussion of digital analysis, files recovered and tools used. Excellent discussion of issues raised.
Individual Report	(0.0 – 1.0) Individual contributions not outlined. Very limited discussion of challenges or success. No mitigations covered. No references and citations.	(1.0 – 3.0) Average individual contributions outlined. Limited discussion of challenges or success. Some mitigations covered. Some references and citations used.	(3.0 – 5.0) Excellent individual contributions outlined. Excellent discussion of challenges or success. Elaborated and addressed mitigations to a high level. References and citations used effectively and formatted correctly.

End of Coursework2